Granite Tait · Oct 28, 2026

Terra

A custom AI model that **moves AI work off the grid's rush hours.**

Scan now. It's live.

terra.granitetait.com

github.com/granman3/terra

BUILT ONLY ON PUBLIC GOVERNMENT DATA

33 million

data points, every hour since January 2022, from **50** US power regions.

33 million data points → 1.04 million examples → 5 training runs, 3 designs → 1 custom model

EIA-930
hourly demand, forecasts and fuel mix for every US grid · eia.gov

EIA-860M
every power plant planned in Utah · eia.gov

Weather forecasts
NOAA and other national models, as forecast at the time · Open-Meteo

CAISO OASIS
Utah's real-time wholesale price, hourly · oasis.caiso.com

Utah PSC
Rocky Mountain Power tariffs and demand-response prices · psc.utah.gov

EIA prices
wholesale power at 5 hubs + Henry Hub gas · eia.gov

`data/train/checks_v2.json`

THE PROBLEM

$1.1 trillion

what US power companies plan to spend on the grid in **just five years**, 2025–29, the most ever. Peak demand is outrunning the grid, and **data centers are the main new driver of that demand.**

DEMAND

+224 GW

new US peak demand in 10 years. 13 of 23 regions now face a risk of running short.

AI'S SHARE

4.4% → 12%

of all US electricity going to data centers, 2023 → 2028

WATER

211B gal

of water a year just to make data centers' power, 12× what they use on site

WASTED FUEL

+40%

more gas per kWh burned by the peaker plants that cover the worst hours

> Why it's so hard: **every load is treated as fixed**, so the whole grid gets built for its worst few hours. **Terra makes the biggest new load move.**

EEI 2025 · NERC 2025 LTRA · DOE/LBNL Dec 2024 (water, 2023) · EIA heat rates: peaker ~10,000 vs 7,146 Btu/kWh

THE PROBLEM, IN UTAH

Utah builds power plants for its worst hour. AI is making that hour worse.

10 GW

the largest data center asking Utah for power. Another was told: not until 2031.

$16.4B

one year of standby plants on the biggest US grid (PJM), and still 6.8 GW short. Building alone isn't keeping up.

48

data centers already in Utah, up to a couple million gallons of water a day each, next to a shrinking Great Salt Lake

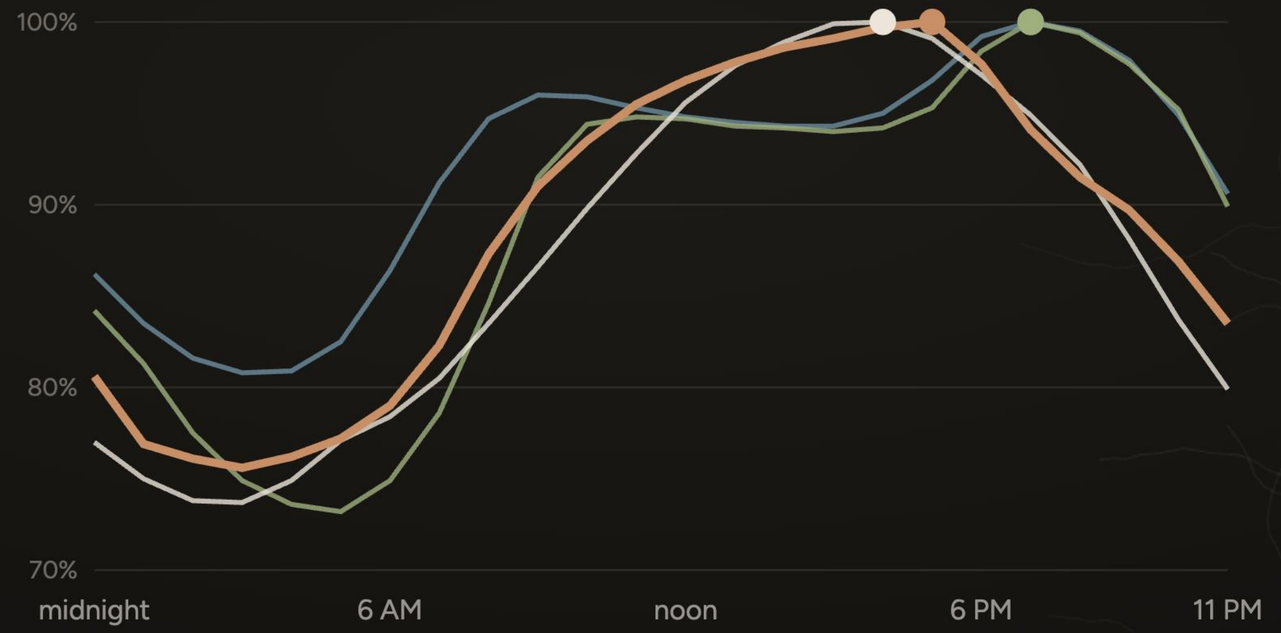
In Utah, the top 10% of the grid, **972 MW**, was needed for only **153 hours** in 2025. That's 1.7% of the year. **Those hours are Terra's target.**

`972,000 kW × $728–1,969 per kW of new gas peaker = $0.7–1.9 billion of plants that aren't needed 98% of the year`

Lehi Free Press, Feb 2026 · PJM 2028/29 auction · Deseret News, Aug 2026 · EIA-930, PACE 2025, hours ≥ 90% of peak · GridLab 2025

THE SOLUTION

Power plants can't move. AI work can.



● Texas peaks 4 PM ● Utah peaks 5 PM ● California peaks 7 PM ● Pacific NW peaks 7 PM

Average demand through the day, 2026, each grid vs its own high (Utah time)

85%

of Utah's rush hours in 2026, California, the Pacific NW or Texas had room to spare

1.14 → 0.04

rush hours a 4-hour AI job runs into: start now in Utah vs Terra

Why AI work: 42% of AI chip time is training and batch work that can wait hours. Nobody notices if it runs in Oregon at 9 PM.

EIA-930, 2026, rush hour = demand ≥ 95% of that day's high · out/v2/metrics.json, held-out 2026 · Alibaba 2026 trace, 155,410 GPUs

WHY I PICKED THIS SOLUTION

Three ways to get through the worst hour. Moving the load costs nothing to build.

BUILD MORE PLANTS

Peakers

$728–1,969 / kW

Years to build. Burns ~40% more fuel per kWh, then sits idle 98% of the year.

STORE THE ENERGY

Batteries

Torus Pulse

Covers the hours the load can't move. Terra moves what can, so Pulse covers the rest.

MOVE THE LOAD

Terra

$0 to build

Software on chips already plugged in. Uses grid room that already exists, in another region or a later hour.

98 GW of new data centers fit on **today's US grid, with no new plants**, if they flex just 0.5% of the time, about 2 hours at a stretch. Terra is how they flex.

GridLab, Sept 2025, new peaker costs · EIA heat rates · Duke Nicholas Institute, "Rethinking Load Growth," Feb 2025 (76-126 GW at 0.25-1%)

WHAT I BUILT

A custom AI model, made for one job. Trained on my own hardware.

EVERY INPUT, EVERY HOUR

- Grid demand + the utility's forecast
- Fuel mix and carbon
- Weather forecasts
- Sun position
- Holidays and calendar
- Power and gas prices

492 numbers per decision · **50** regions

→

THE TERRA MODEL

561K settings

Designed, trained and tested by me at home on one RTX 4080: 5 training runs, 3 designs, the best one kept.

→

DECISION → TORUS LASSO

Where, when, how hard

for every AI job, in a fraction of a millisecond. A dispatchable asset for Lasso.

33 million
public data points

1.04 million
examples, 2022–26

29%
fewer mistakes than the utility's forecast

20 joules
a full year of decisions, all 50 regions

> Why custom: no off-the-shelf model forecasts every US grid's rush hour a day ahead, cheaply enough to **save more energy than it spends.**

held-out 2026 data, Brier score vs EIA's calibrated day-ahead forecast · out/v2/metrics.json · out/mac/tiny_efficiency.json

THE MATH INSIDE TERRA

One equation picks where and when AI work runs.

$$(r^*,\ s^*)\ =\ \arg\ \min_{\substack{r\,\in\,\text{regions}\\ s\,\in\,\{0,3,6,12\},\ s+4\,\le\,D}}\ \sum_{h=s}^{s+3}\left(\hat{p}_{r,h}\ +\ \frac{g_{r,h}}{4000}\right)$$

Pick the region **r** and start hour **s** (now, +3, +6 or +12 h, within the job's deadline **D**) where the job's 4 hours carry the **least chance of rush hour**, carbon as the tiebreak.

THE FORECAST

$$\hat{p}_{r,h}=\sigma\left(f_\theta(x_{r,t})_h\,/\,T\right),\quad x\in\mathbb{R}^{492}$$

492 inputs per region per hour in, chance of rush hour for each of the next 24 h out. **561K** learned settings θ, calibrated by T.

HOW IT LEARNED

$$\min_\theta\ \mathrm{BCE}(\hat{p},y)+4\,\mathrm{MSE}_{\text{share}}+4\,\mathrm{MSE}_{\mathrm{CO_2}}$$

Rush hour, fuel share and carbon learned together, on **755K** examples from 2022–25.

HOW IT'S GRADED

$$\text{Brier}=\frac{1}{N}\sum_i(\hat{p}_i-y_i)^2$$

On 2026 hours it never saw: **0.050** vs **0.071** for the utility's own forecast.

> Result: a 4-hour AI job's rush-hour exposure drops from **1.14 hours** to **0.04**.

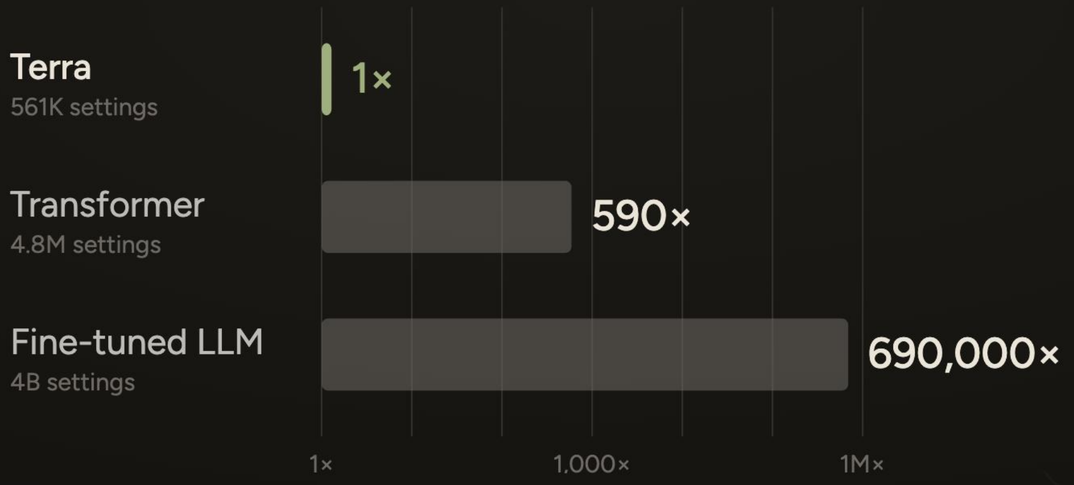
router/src/decide.ts (the decision), train/train.py (the model and its loss) · out/v2/metrics.json, 1,162 held-out 2026 decisions

WHY A SMALL CUSTOM MODEL

Bigger AI wasn't better. I tested it.

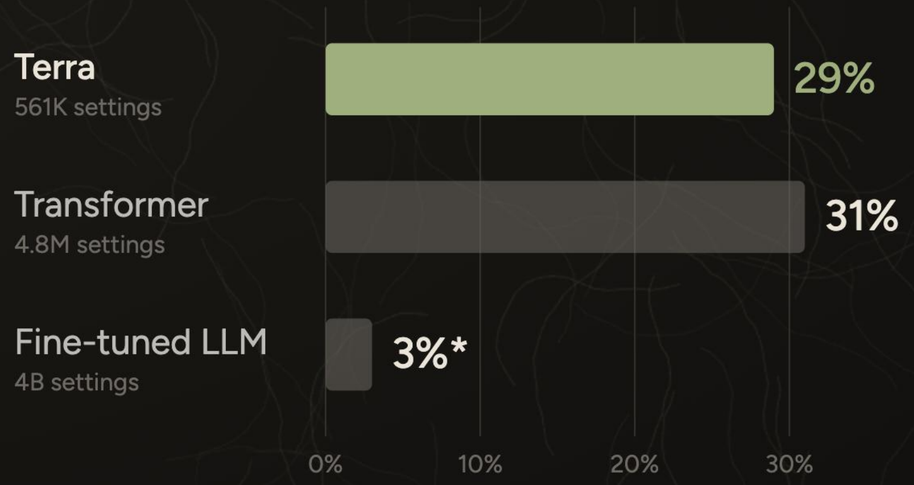
Energy per decision

times Terra's, log scale (each step is 10×)



Fewer mistakes than the utility's forecast

higher is better



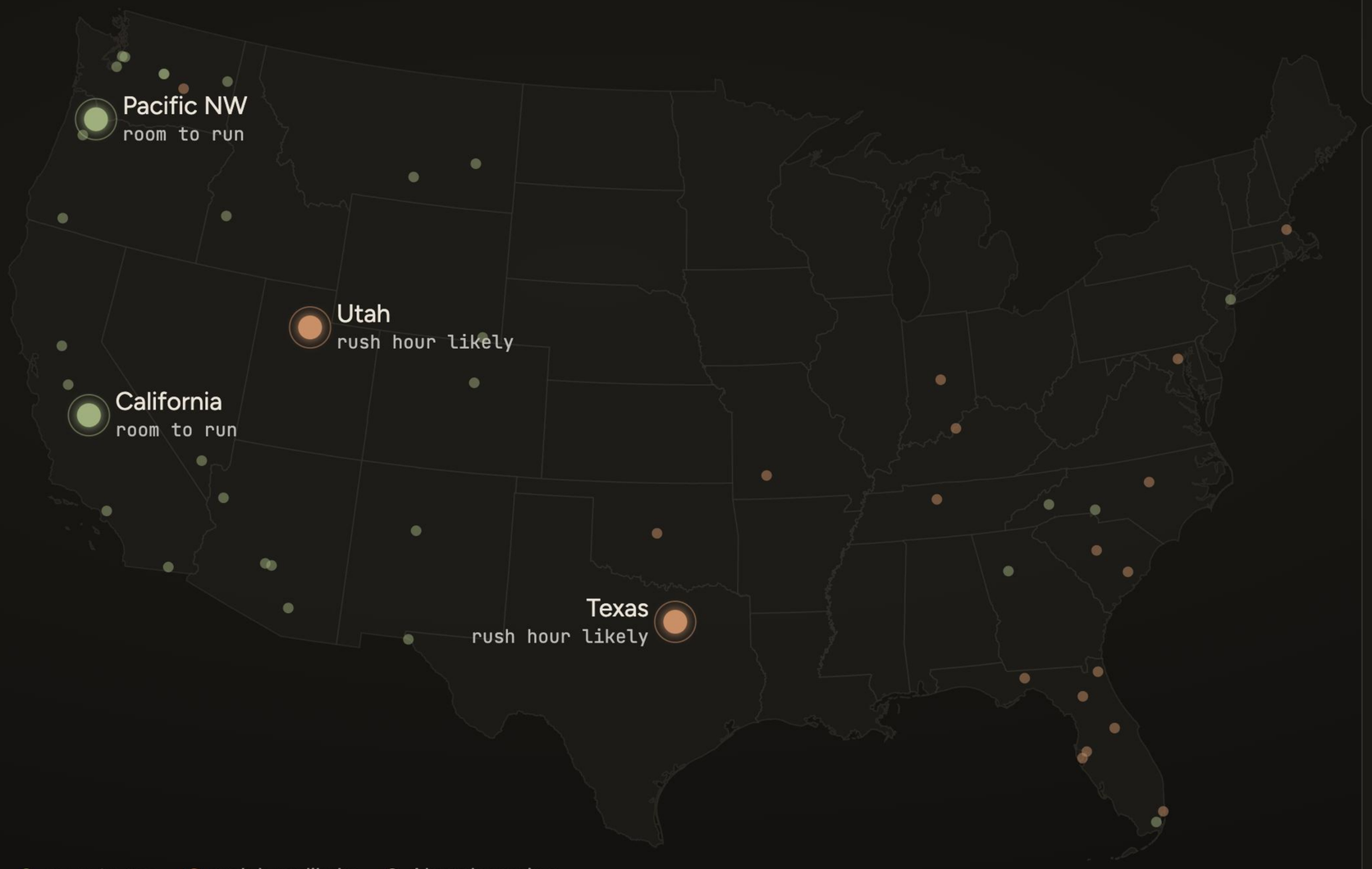
> The 4-billion-parameter AI model used **690,000× the energy** and was barely better than the utility. Terra is **small on purpose: we don't spend energy to save energy.**

all three trained by me on the same RTX 4080 · energy: Transformer vs Terra on one laptop CPU core, LLM vs Terra total chip power on the same MacBook · *LLM accuracy from the earlier 4-region test

Terra LIVE

Sun 3 PM in Utah · power use at 100% of today's high · RUSH HOUR

50 regions forecast · 20 expecting rush hour



room to run · rush hour likely · AI work moving

RIGHT NOW

AI model training → Pacific NW, in 12 hours

Utah rush-hour chance 96%

WHAT UTAH SAVES

$10M a year

per **100 MW** of AI work moved off Utah's rush hours

THIS WEEK · ONE SIMULATED SITE

Power moved off the rush	**736 kWh**
AI jobs moved	**64 of 64**
Saved so far	**$63.25**

10 / 11

Real EIA grid data, Oct 19–25, 2026, unseen in training · real Utah prices (CAISO) · the data center and its jobs are simulated

WHY IT MATTERS: UTAH, EVERY GRID, AND TORUS

TORUS SPIN

Milliseconds

Flywheel catches the fast swings.

TORUS PULSE

Hours

Battery covers the long rush hour.

TERRA

The load itself

Moves the demand: to a region with room, or a later hour. A third asset for Lasso, nothing new to build.

UTAH

$10M a year

for every 100 MW of AI work that can wait: 100,000 kW × $100 per kW a year, Rocky Mountain Power's own price

EVERY GRID

98 GW

of new AI load today's US grid can take with no new plants, once it flexes off the peaks

POWER SPIKES

813 → 16 kW

a 1 MW AI site's one-second jump with staggered starts. Less for Spin to catch.

HOW WE GET THERE

Built

Forecast the rush, a day ahead

50 regions. Utah: 80% of rush hours caught vs 69% for the utility.

Next

Plug into Lasso

Pilot with one Utah AI site on Rocky Mountain Power's demand-response program.

Then

Every GPU on the grid

Utah's regulators have a docket open on a flexible large-load tariff (PSC 24-035-43).

Wattsmart Business demand response, Rocky Mountain Power · Utah Code 54-26 (SB 132, 2025), PSC docket 24-035-43 · rush hours caught: held-out 2026 data